Regularized Multivariate Two-way Functional Principal Component Analysis

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Outline

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2. **A SVD Approach for Regularized Multivariate FPCA**
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Introduction

Functional Data Analysis:

- Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.
- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.

Regularized MFPCA

Two-way Regularized MFPCA

Conclusion & Future Work